A Literature-Implied Continuous Selection for Bounded $C^{1,\omega}$ Whitney Jets

Agent lane 8 note for arXiv:1912.13317

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Status

This is a literature-implied answer to the weaker continuity alternative in Remark 4.3(2) of Azagra–Mudarra [1]. The decisive external input is the classical Bartle–Graves theorem: a continuous surjective linear map between Banach spaces admits a continuous nonlinear right inverse [2, 3]. The observation is that the bounded $C^{1,\omega}$ trace problem in [1] is exactly such a quotient-map situation.

Source Question

The source question appears on page 26 of the arXiv PDF, in Remark 4.3(2) of [1].

> 26 DANIEL AZAGRA AND CARLOS MUDARRA
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> (2) It would be interesting to know whether one can improve Theorems 1.5 and 4.1 to find an extension operator with the additional property that
>
> $$\lim_{n\to\infty} A(\mathcal{E}(f_n, G_n) - \mathcal{E}(f, G), \nabla\mathcal{E}(f_n - f, G_n) - \nabla\mathcal{E}(f, G)) = 0$$
>
> whenever $A(f_n - f, G_n - G) \to 0$, or at least such that $\mathcal{E}$ be continuous from the normed space $(\mathcal{J}_b^{1,\omega}(E), \rho)$ into $(C_b^{1,\omega}(X), \|\cdot\|_{1,\omega,b})$.

In words, the remark asks whether the bounded extension results can be improved to obtain an extension operator with stronger A-seminorm continuity, or at least an operator continuous from

$$(\mathcal{J}_b^{1,\omega}(E), \rho) \quad \text{into} \quad (C_b^{1,\omega}(X), \|\cdot\|_{1,\omega,b}).$$

Identification

The paper already proves, in Theorem 4.1, that every bounded Whitney jet $(f, G) \in \mathcal{J}_b^{1,\omega}(E)$ admits an extension $F \in C_b^{1,\omega}(X)$, with

$$\|F\|_{1,\omega,b} \leq C\,\rho(f, G).$$

The remaining point is not another extension formula. It is a selection problem for the linear trace operator. Once the trace space is viewed with the norm ρ, Theorem 4.1 says precisely that this trace operator is onto and has quotient norm equivalent to ρ. Bartle–Graves then gives a continuous nonlinear choice of preimage.

Theorem

Theorem 1. *Let X be a Hilbert space, let $E \subset X$, and let ω be under the hypotheses of Azagra–Mudarra's bounded extension theorem. Then there is a nonlinear extension operator*

$$S : (\mathcal{J}_b^{1,\omega}(E), \rho) \longrightarrow (C_b^{1,\omega}(X), \|\cdot\|_{1,\omega,b})$$

such that

$$\big(S(f,G), \nabla S(f,G)\big)|_E = (f,G)$$

for every $(f,G) \in \mathcal{J}_b^{1,\omega}(E)$, and S is continuous.

Consequently, if $\rho((f_n,G_n) - (f,G)) \to 0$, then

$$\|S(f_n,G_n) - S(f,G)\|_{1,\omega,b} \to 0.$$

In particular,

$$M_\omega\big(\nabla S(f_n,G_n) - \nabla S(f,G)\big) \to 0$$

and hence

$$A\big(S(f_n,G_n) - S(f,G), \nabla S(f_n,G_n) - \nabla S(f,G)\big) \to 0.$$

Proof. Let

$$B = (C_b^{1,\omega}(X), \|\cdot\|_{1,\omega,b})$$

and define the trace operator

$$T : B \to \mathcal{J}_b^{1,\omega}(E), \qquad T(F) = (F|_E, \nabla F|_E).$$

This map is linear. It is bounded because

$$\|F|_E\|_\infty \le \|F\|_\infty, \qquad \|\nabla F|_E\|_\infty \le \|\nabla F\|_\infty,$$

and, as recorded in the source paper before Corollary 1.6, every $C^{1,\omega}$ extension H of a jet satisfies

$$A(f,G) \le M_\omega(\nabla H).$$

Thus

$$\rho(TF) \le \|F\|_{1,\omega,b}.$$

Theorem 4.1 of [1] gives, for every $(f,G) \in \mathcal{J}_b^{1,\omega}(E)$, an extension $F \in B$ with

$$\|F\|_{1,\omega,b} \le C\rho(f,G).$$

Therefore T is onto. If q denotes the quotient norm induced by T,

$$q(f,G) = \inf\{\|F\|_{1,\omega,b} : T(F) = (f,G)\},$$

then the two preceding estimates give

$$\rho(f,G) \le q(f,G) \le C\rho(f,G).$$

The quotient $B/\ker T$ is a Banach space, and q is its norm under the natural identification with the trace space. Since q and ρ are equivalent, $(\mathcal{J}_b^{1,\omega}(E), \rho)$ is Banach.

We may now apply the Bartle–Graves theorem to the bounded surjective linear operator

$$T : B \to (\mathcal{J}_b^{1,\omega}(E), \rho).$$

It yields a continuous nonlinear map $S : (\mathcal{J}_b^{1,\omega}(E), \rho) \to B$ such that $T \circ S = \mathrm{id}_{\mathcal{J}_b^{1,\omega}(E)}$. This identity is exactly the asserted extension property, and continuity of S gives the norm convergence claim. The last displayed A-convergence follows from the source estimate $A(H, \nabla H) \le M_\omega(\nabla H)$, applied to

$$H = S(f_n, G_n) - S(f, G).$$

□

Scope And Limitations

The result answers the "or at least" part of Remark 4.3(2). It also gives the natural A-seminorm convergence for the difference of selected extensions. It does not show that Azagra–Mudarra's explicit paraconvex-envelope operator has this continuity, nor does it give a new closed formula for a continuous selector. The proof is nonconstructive through Bartle–Graves.

The displayed formula in the source remark appears to contain a typographical mismatch in the gradient term. The conclusion above is stated in the natural form forced by convergence in $\|\cdot\|_{1,\omega,b}$, namely for the gradient of the difference of the selected extensions.

Search Notes

Local run indexes were searched for '1912.13317', 'C1omega', '1-jets', 'Whitney', 'continuous extension operator', and 'Bartle-Graves'. Nearby Whitney packets for arXiv:2305.19995 and arXiv:2403.14317 concern different questions. Web searches on 2026-06-19 found the source arXiv page and general Bartle–Graves references, but no paper explicitly identifying the above implication for Remark 4.3(2). Accordingly this packet is recorded as a literature-implied answer, not as an original construction.

References

[1] D. Azagra and C. Mudarra, *$C^{1,\omega}$ extension formulas for 1-jets on Hilbert spaces*, arXiv:1912.13317.

[2] R. G. Bartle and L. M. Graves, *Mappings between function spaces*, Transactions of the American Mathematical Society 72 (1952), 400–413.

[3] Y. Benyamini and J. Lindenstrauss, *Geometric Nonlinear Functional Analysis. Vol. 1*, American Mathematical Society Colloquium Publications 48, 2000.